

# Global Renewable Energy in 2025:

## How Far Have We Come?

A data-driven policy briefing on renewable energy across primary energy, electricity, and leading technologies.

**~14%**

**global primary energy from  
renewables**

2024, substitution method

**~30%**

**global electricity from renewables**

2025 global average

**~9,600**

**TWh renewable electricity**

hydro + wind + solar + other

# 75% of global GHG emissions still come from fossil fuels

**75%**

**of global greenhouse gas emissions come from burning fossil fuels for energy**

The energy system remains the core climate challenge.



Air pollution burden

**5M+ premature deaths / year**



System lock-in

**Fossil fuels dominate since the Industrial Revolution**

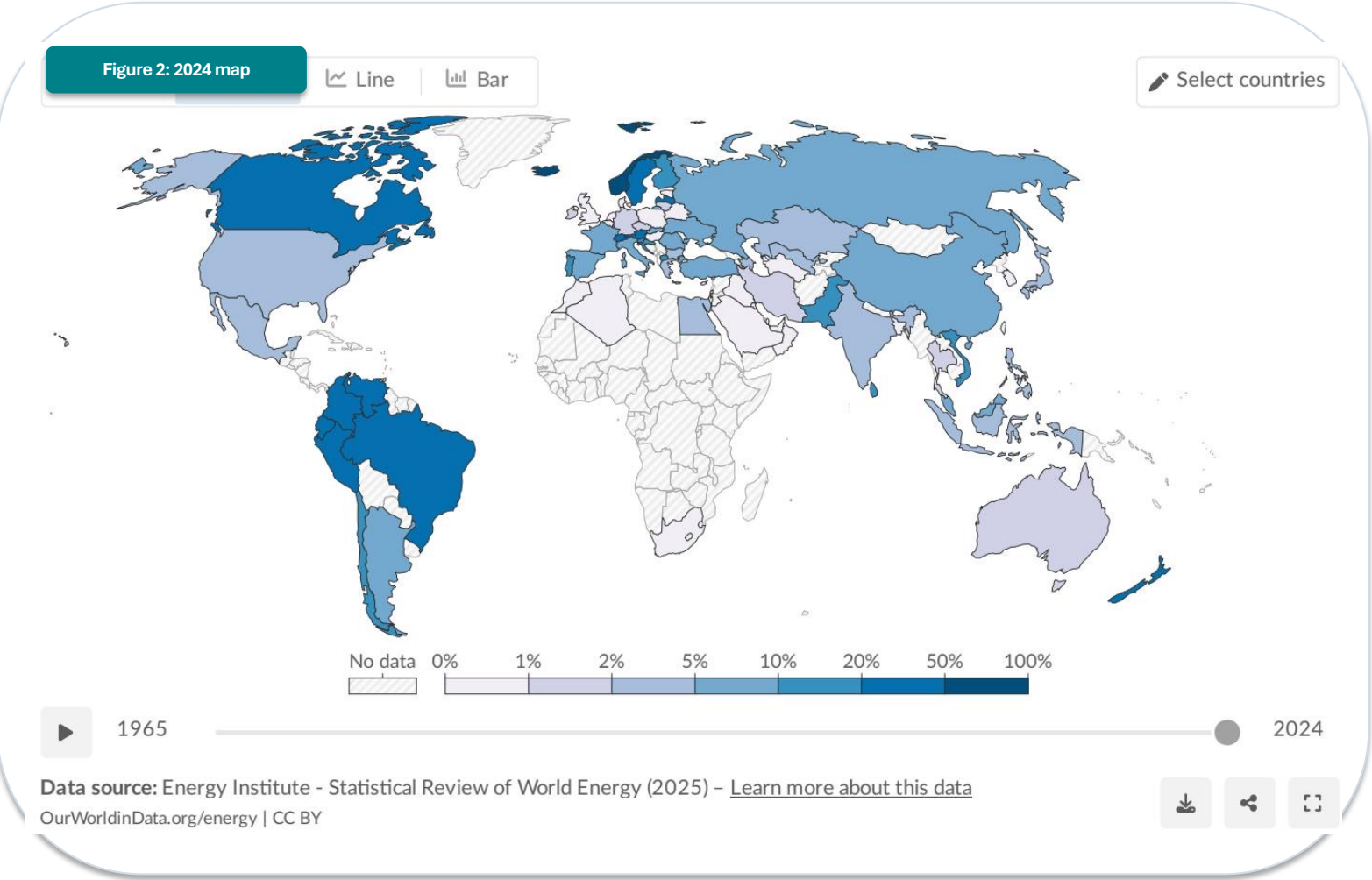


Transition imperative

**Rapid shift to low-carbon energy: renewables + nuclear**

**Modern renewables exclude traditional biomass in lower-income settings.**

# Renewables now supply ~14% of global primary energy



**~14%**

of global primary energy is renewable

about one-seventh

**Lower**

than electricity share

transport + heating remain harder to decarbonize

**<5%**

in some fossil-fuel producers

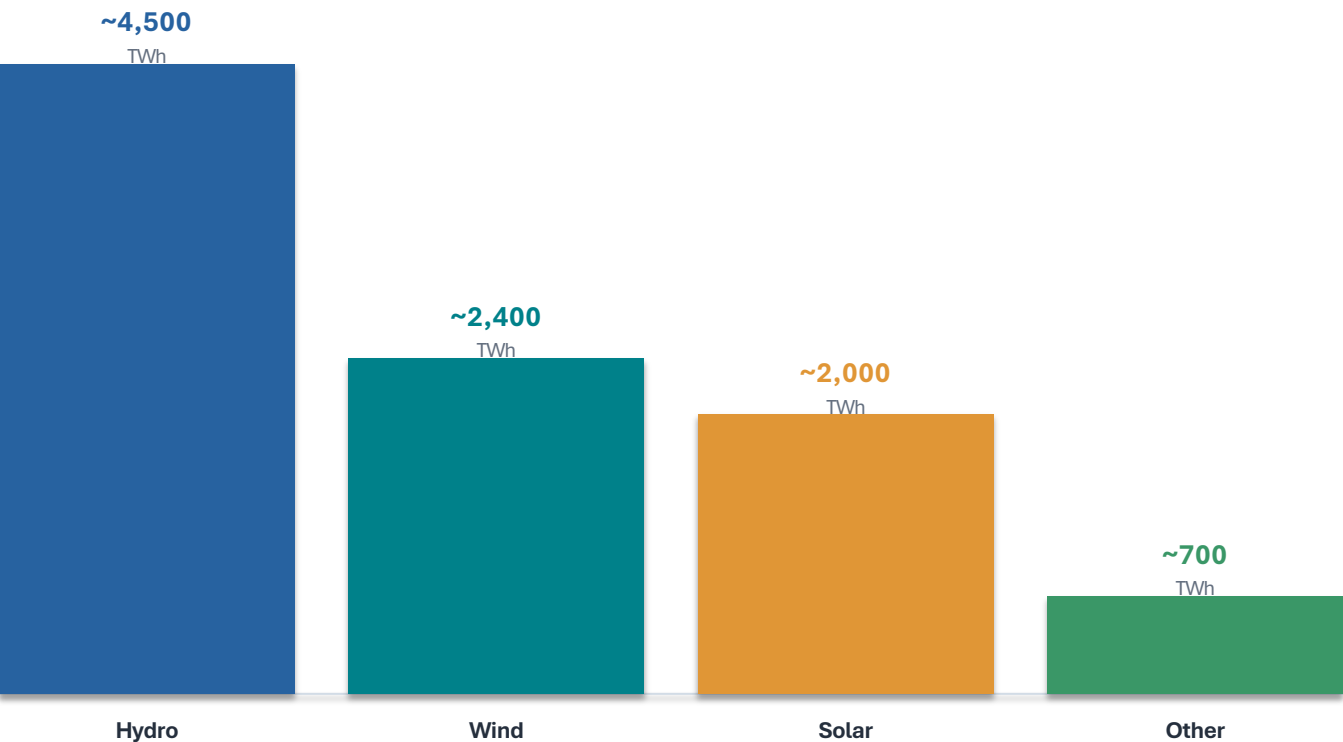
Middle East and parts of Asia

Method note

Primary energy uses the substitution method; it spans electricity, transport, and heating.

# Renewable electricity generation is growing rapidly, led by wind and solar

Approx. global generation by renewable technology, 2024



| Technology       | Generation | Share |
|------------------|------------|-------|
| Hydropower       | ~4,500 TWh | ~47%  |
| Wind             | ~2,400 TWh | ~25%  |
| Solar            | ~2,000 TWh | ~21%  |
| Other renewables | ~700 TWh   | ~7%   |

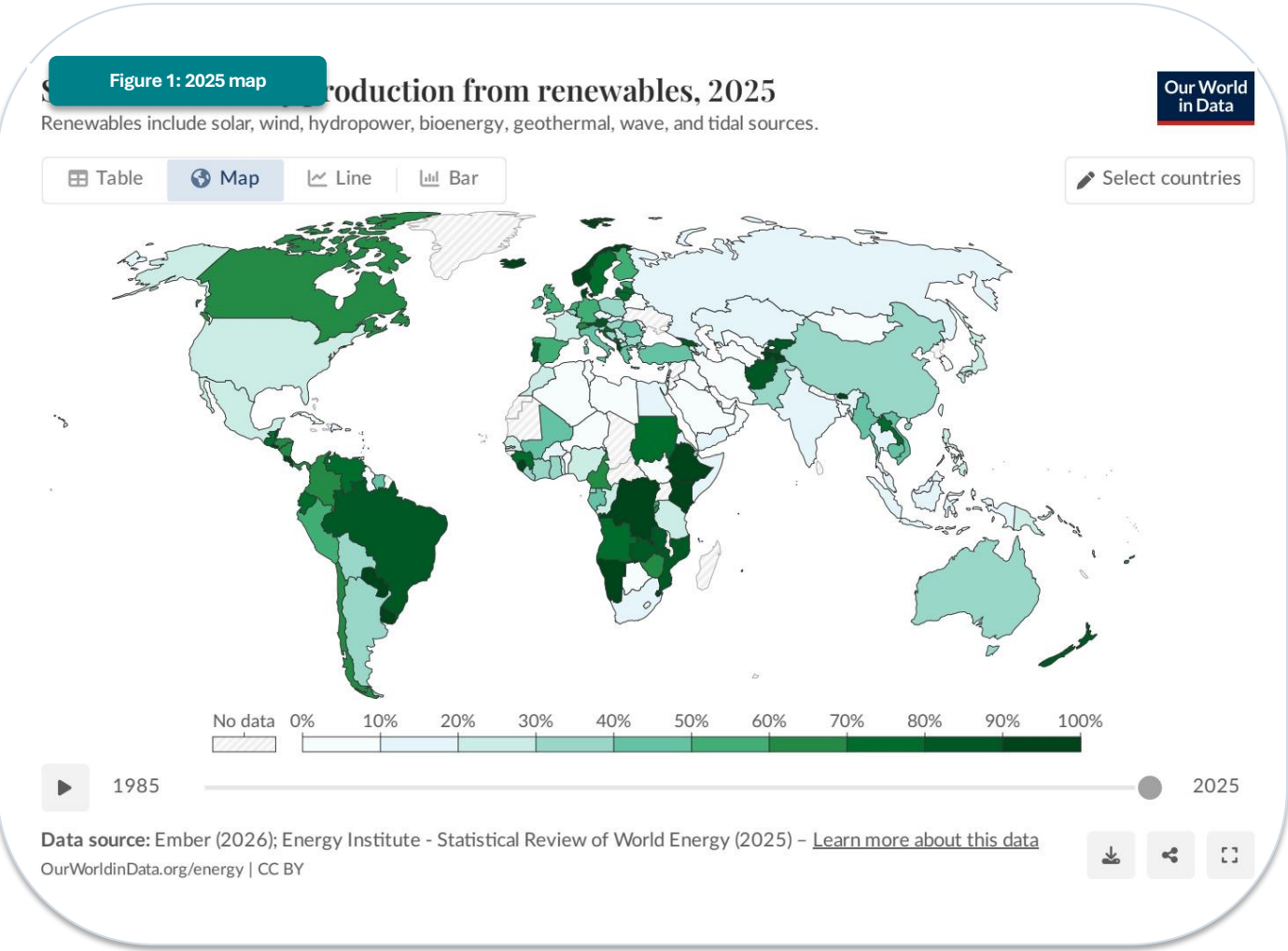
Wind and solar combined now nearly match hydropower output.

Wind and solar growth has accelerated sharply since around 2010.

Hydropower remains largest, but its growth potential is more limited.

Solar is the fastest-growing renewable source, driven by falling costs and policy support.

# Almost one-third of global electricity now comes from renewables



~30%

world average

60–90%+

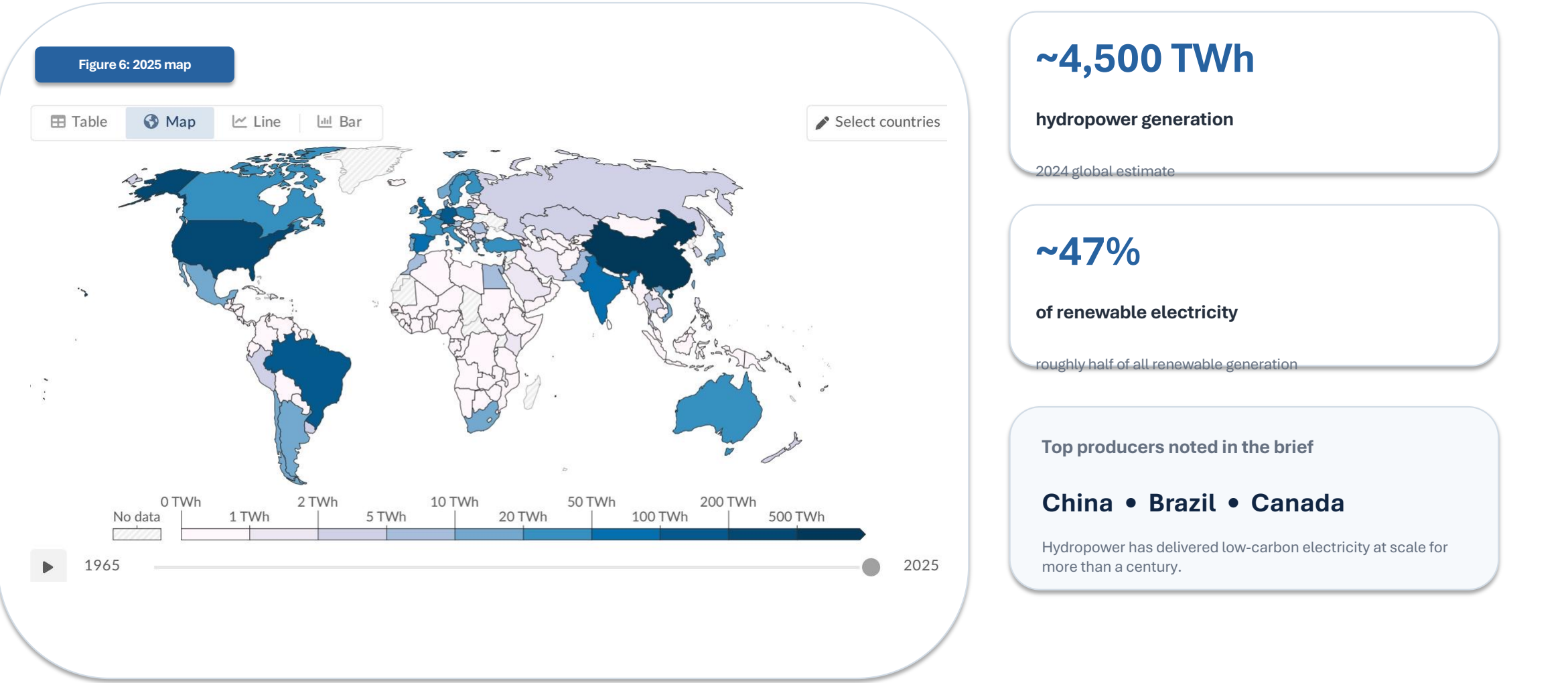
regional leaders

Selected country / region shares

| Country / region | Renewable electricity |
|------------------|-----------------------|
| Norway           | ~98%                  |
| Brazil           | ~85%                  |
| Canada           | ~65%                  |
| China            | ~35%                  |
| India            | ~25%                  |
| Saudi Arabia     | <2%                   |

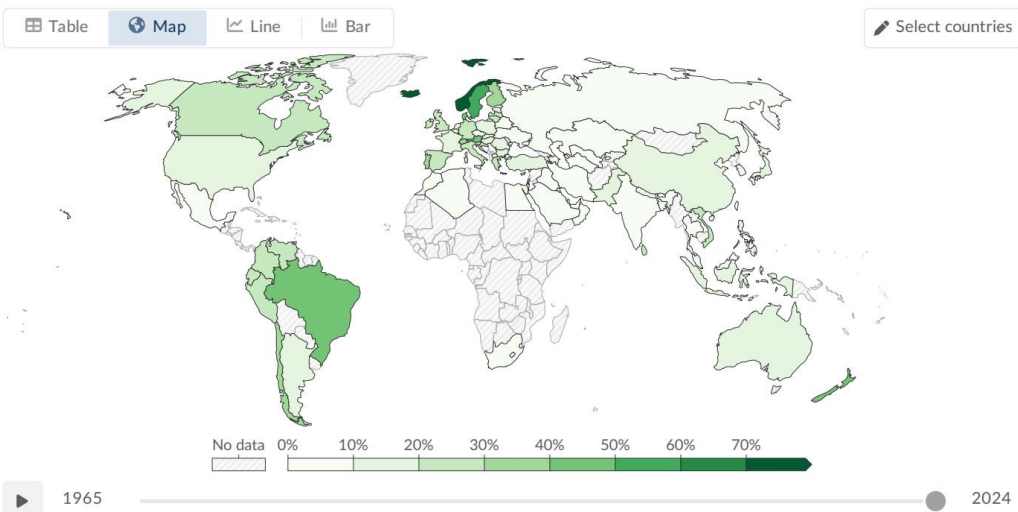
Electricity is the sector where renewables have made the most visible progress.

# Hydropower remains the largest renewable source at ~50% of generation



# Wind and solar are the fastest-growing energy technologies in history

Figure 8: wind map



## Generation and share of renewable electricity

### Wind

**~2,400 TWh**

generation

**~25%**

of renewable electricity

### Solar

**~2,000 TWh**

generation

**~21%**

of renewable electricity

**Together, wind + solar now nearly match hydropower output.**

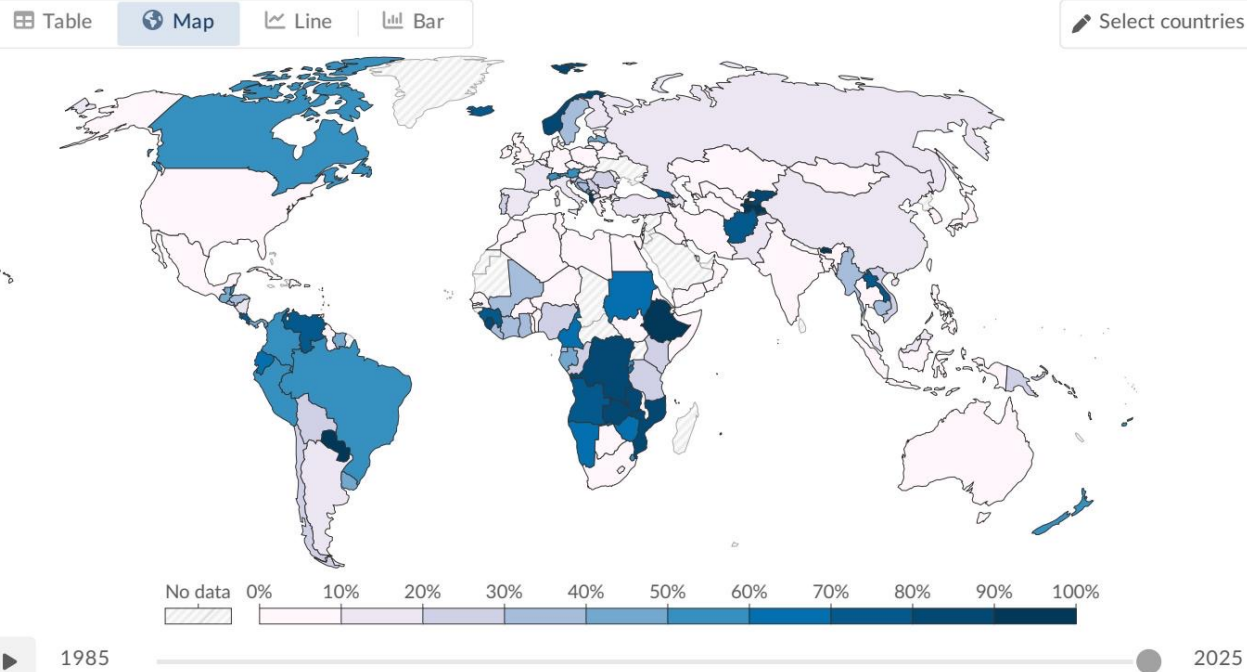
Wind is now a major contributor in Europe, China, and the United States.

Solar has increased dramatically since 2010, supported by falling costs and policy support.

The available source data compares generation and shares—not installed capacity.

# Stark regional gaps: Sub-Saharan Africa lags while Scandinavia and Latin America lead

Figure 3: hydropower share



The uneven transition reflects geography, policy, investment, and legacy systems.

## Leaders

Scandinavia, Brazil, Canada: 65–98% renewable electricity

## Wind + solar adopters

Denmark and Germany have invested heavily in wind and solar

## Low-share regions

Several African and Middle Eastern countries remain below 10%

## Fossil dependence

Many Middle Eastern nations remain almost entirely fossil-fuel dependent



# Key takeaways: renewables are growing fast, but the transition must accelerate

## ~14% primary energy

Renewables supply about one-seventh of total energy across electricity, transport, and heating.

## ~30% electricity

Electricity is further ahead, with almost one-third generated from renewable sources.

## Hydro still dominant

Hydropower generates roughly half of renewable electricity, led by China, Brazil, and Canada.

## Wind + solar catching up

Together they now produce nearly as much electricity as hydropower and are growing much faster.

## Regional gaps persist

Leaders exceed 60–90%, while parts of the Middle East and Sub-Saharan Africa remain below 10%.

**Strategic implication: accelerate renewable deployment beyond electricity—especially transport and heating—to close the primary-energy gap.**